

A PROJECT REPORT
ON
**Customer Churn Prediction in Banking Using XGBoost
and Random Forest**

project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

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By

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Abstract

This project develops a supervised machine learning system for predicting customer churn in the banking sector. Customer profile, account activity and financial attributes are prepared through data inspection, cleaning, categorical encoding and exploratory analysis. Random Forest and XGBoost are used as the two tree-based classification models. The system predicts whether a customer is likely to churn (Exited = 1) or remain with the bank (Exited = 0), and model performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC and a confusion matrix.

Keywords: Customer Churn, Banking Analytics, Random Forest, XGBoost, Binary Classification, Churn Prediction

1. Introduction

1.1 Purpose of the Project

Customer churn in banking occurs when a customer closes an account or stops using the bank's services. Banks manage large customer populations with different demographic profiles, balances, product usage, activity levels and credit characteristics. Identifying customers who may leave is difficult when decisions rely only on manual review or fixed rules. A machine learning approach can learn patterns from historical customer records and produce a consistent churn prediction for new customers.

The project is intended for banking analytics and customer-retention analysis. Relevant attributes in the supplied project include CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember and EstimatedSalary. The output is designed to support early identification of customers who may be at higher churn risk.

1.2 Problem Statement

The problem is formulated as a supervised binary classification task. Given a customer's demographic, account and activity-related attributes, the system predicts whether the customer will churn or stay with the bank. The target variable is Exited, where Exited = 1 represents a churned customer and Exited = 0 represents a retained customer. The expected output is a Churn / No Churn classification and, where probability prediction is enabled, an estimated churn probability.

1.3 Existing Methods and Disadvantages

Traditional churn analysis may use manual customer review, fixed business rules, threshold-based alerts or simple statistical comparisons. These approaches can be difficult to scale when the number of customers and attributes becomes large. Fixed rules can also miss interactions among variables such as age, balance, activity status, geography and number of products.

Tree-based ensemble methods provide a data-driven alternative. Random Forest aggregates many decision trees and can capture nonlinear relationships, while XGBoost builds boosted trees sequentially to improve predictive performance. Both methods still require careful preprocessing, leakage prevention, validation and evaluation; class imbalance and model interpretability should also be considered when applying predictions in a banking setting.

1.4 Proposed System

The proposed system follows the pipeline: banking churn dataset → data inspection → cleaning and duplicate checking → categorical encoding → feature preparation → train/test split → Random Forest and XGBoost training → prediction → evaluation. Identifier fields such as RowNumber, CustomerId and Surname are removed when present because they are not intended as predictive customer-behaviour features. The same training and testing split is used for both models so that the comparison is consistent. The expected contribution is a reproducible churn-classification workflow for banking customer analytics.

1.5 Research Objectives (at least 4)

1. Prepare and understand a suitable banking customer churn dataset.
2. Clean the data and prepare numerical and categorical features.
3. Perform exploratory data analysis to identify churn-related patterns.
4. Train Random Forest and XGBoost classification models.
5. Evaluate the models using accuracy, precision, recall, F1-score and ROC-AUC.
6. Generate a final Churn / No Churn prediction with an associated churn probability.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

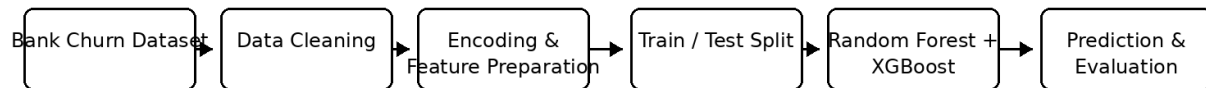
S.No.	Title	Reference / Source status	Problem or gap addressed	Objective of the Research	Focus of the paper / methods	Inference	Recommendations for future research
1	Bank Customer Churn Prediction	Verified paper details not supplied in project files	Early identification of customers likely to leave	Predict churn from customer/account attributes	Supervised classification; customer demographics and account variables	ML can convert historical customer records into churn-risk predictions	Validate on external banking datasets
2	Customer Churn and Retention Analytics	Verified paper details not supplied in project files	Manual or rule-based retention analysis does not scale well	Support data-driven retention analysis	Classification and customer-behaviour analysis	Customer behaviour variables can be used for retention analytics	Include richer transaction and engagement data
3	Tree-Based Models for Churn Classification	Verified paper details not supplied in project files	Complex interactions between churn factors	Compare ensemble tree classifiers	Random Forest / boosted trees	Tree ensembles can model nonlinear relationships	Tune models with cross-validation
4	Random Forest for Banking Churn	Verified paper details not supplied in project files	Need robust classification across mixed customer features	Develop a robust churn classifier	Random Forest; ensemble decision trees	Multiple trees provide aggregated classification and feature importance	Study explainability and calibration
5	XGBoost for Banking Churn	Verified paper details not supplied in project files	Need efficient boosted-tree prediction	Evaluate gradient boosting for churn	XGBoost; boosted decision trees	Boosting focuses later trees on previous errors	Evaluate tuning and probability calibration
6	Class Imbalance in Churn Prediction	Verified paper details not supplied in project files	Churn classes may be imbalanced	Improve detection of minority churn cases	Class weighting, stratified splitting and threshold analysis	Accuracy alone may hide minority-class errors	Report precision, recall and F1 with class balance
7	Feature Analysis for Customer Churn	Verified paper details not supplied in project files	Redundant identifiers can introduce noise or leakage	Prepare valid predictive features	Remove IDs; encode categorical variables; inspect correlations	Feature validity is essential before training	Test feature selection and explainability methods
8	Explainable Banking Churn Prediction	Verified paper details not supplied in project files	Stakeholders need interpretable risk information	Explain important factors behind predictions	Feature importance; future SHAP-style analysis	Tree models can provide feature-importance information	Add customer-level explanations
9	Comparative Churn Model Evaluation	Verified paper details not supplied in project files	Different models can produce different error profiles	Compare models on the same held-out data	Accuracy, precision, recall, F1, ROC-AUC and confusion matrix	Multiple metrics provide a fuller view of classification behaviour	Use cross-validation and external validation
10	Deployment of	Verified paper details	Offline models	Move churn	API/cloud application and	Deployment	Evaluate model drift and

	Banking Churn Models	not supplied in project files	may not directly support real-time use	prediction toward practical deployment	monitoring as future direction	requires monitoring, recalibration and data-drift checks	real-time scoring
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3. Methodology

3.1 Proposed System Flow Diagram

Proposed system flow for the banking customer churn prediction project:



Dataset → Data Cleaning → Encoding & Feature Preparation → Train/Test Split → Random Forest + XGBoost
→ Prediction & Evaluation

3.2 Dataset Description

Dataset: bank_churn.csv, as specified in the project presentation. The dataset represents banking customers and contains profile and account-related attributes. Typical variables include CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember and EstimatedSalary, with Exited as the churn target. Exited = 1 denotes churn and Exited = 0 denotes retention. The exact number of records, final feature count and class distribution must be taken from the executed bank_churn.csv file because the supplied presentation does not report those final numeric values.

3.3 Experimental Setup

Experimental setup: Python-based machine learning environment using Pandas and NumPy for data handling, Matplotlib for visualization, scikit-learn for train/test splitting, Random Forest and evaluation metrics, and XGBoost for gradient-boosted classification. The supplied project code loads the dataset with `pandas.read_csv("bank_churn.csv")`. The exact hardware, operating system, Python version and package versions should be recorded from the environment used for the final execution.

3.4 Data Preprocessing

- Data inspection and schema validation
- Duplicate removal
- Missing-value analysis and imputation
- Outlier analysis/treatment where justified
- Categorical encoding
- Numerical scaling/standardization where required
- Target-label preparation
- Train/test split and prevention of data leakage

3.5 Exploratory Data Analysis

EDA for this project focuses on churn-class distribution, churn versus age, churn versus geography, churn versus active membership, churn versus number of products and correlation analysis among numerical variables. These plots help identify patterns in the banking customer data before model training. The final report should include the actual plots generated from bank_churn.csv.

3.6 Feature Engineering and Feature Selection (if applicable)

Feature preparation uses the available customer attributes after removing identifier columns such as RowNumber, CustomerId and Surname when present. Geography and Gender are encoded as numerical features. Numerical attributes are retained in their usable form for the tree-based models. Feature selection is guided by data validity and leakage prevention; the target variable Exited is not used to construct predictive features.

3.7 Model Building and Training

Two supervised tree-based classifiers are trained. Random Forest uses `n_estimators=200`, `random_state=42` and `class_weight="balanced"` in the supplied project configuration. XGBoost uses `n_estimators=200`, `max_depth=4`, `learning_rate=0.05`, `random_state=42` and `eval_metric="logloss"`. Both models are trained on the same training set and evaluated on the held-out test set. Random Forest combines many decision trees, while XGBoost builds boosted trees sequentially.

3.8 Hyperparameter Tuning(if applicable)

Hyperparameter tuning, if performed, should use validation data or cross-validation rather than the final test set. Relevant parameters include the number of trees, tree depth and learning rate. The supplied project presentation provides baseline model parameters but does not provide completed tuning results, so the final selected parameters should be recorded from the actual experiment.

3.9 Model Evaluation

Because customer churn is a binary classification problem, the project uses Accuracy, Precision, Recall, F1-score and ROC-AUC. A confusion matrix is used to inspect true positives, true negatives, false positives and false negatives. Precision measures the correctness of predicted churn cases, recall measures how many actual churn cases are detected, F1-score balances precision and recall, and ROC-AUC measures separation between churn and non-churn cases across thresholds.

4. Results and Discussion

4.1 Preprocessing Results

The supplied project materials specify inspection of missing values, duplicate records, data dimensions and target distribution, followed by removal of identifier fields and encoding of categorical variables. However, the final numeric counts are not included in the supplied presentation. These values should be copied from the actual execution of `bank_churn.csv` rather than estimated.

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Random Forest	Actual run required	Actual run required	Actual run required	Actual run required	Actual run required
XGBoost	Actual run required	Actual run required	Actual run required	Actual run required	Actual run required
Test-set comparison	Use measured values	Use measured values	Use measured values	Use measured values	Use measured values

4.3 Visual Results and discussion

For this classification project, the relevant visual results are: churn-class distribution, EDA plots, confusion matrices for Random Forest and XGBoost, ROC curves, and feature-importance plots. These figures should be

generated from the actual model run and inserted here. Prediction-vs-actual and residual plots are not primary evaluation plots for this binary classification task.

The final discussion should compare Random Forest and XGBoost using the measured test-set Accuracy, Precision, Recall, F1-score and ROC-AUC values. It should also discuss false positives and false negatives from the confusion matrices, generalization to unseen customers and the practical meaning of churn probabilities. The supplied project presentation does not contain completed numerical test results, so no model performance value is assumed in this report.

5. Conclusion

This project presents a machine learning approach for predicting customer churn in banking using Random Forest and XGBoost. Customer profile and account-related attributes are inspected, cleaned, encoded and analyzed before model training. Both classifiers generate Churn / No Churn predictions and can provide churn probabilities. The final model comparison must be based on the actual test-set metrics produced from bank_churn.csv. The resulting workflow can support further banking analytics and customer-retention analysis.

6. Future Work

- Use larger and more diverse datasets.
- Evaluate additional algorithms and ensemble methods.
- Perform deeper hyperparameter optimization.
- Investigate explainable AI techniques.
- Evaluate robustness and model drift where applicable.
- Develop a deployable real-time/cloud/edge application.
- Validate the approach on external datasets.

7. References

References should contain the verified papers and technical sources actually used in the final report. The supplied project PPT contains no bibliographic records or DOI information, so this report does not invent author names or DOI numbers. Add the verified recent literature sources selected for the literature-survey table before final academic submission.

Signature of the Guide

Course Instructor